Excelerate AI-Powered Virtual Internship (May - June 2025)

Project Title: AI-Powered Learner Engagement & Churn Analysis

Team Name: Team E

Team Members : Diya Kharel (Team Lead), Iqra Shaikh (Project Lead), Faith Odhe (Team Member)

Internship Overview

Duration: 4 Weeks of immersive learning and project execution

Main Goal: Use data to reduce student drop-offs and improve engagement

Weekly Progress Overview -

Week 1: Data cleaning & feature engineering

Week 2: Exploratory data analysis

Week 3: Predictive modeling of churn

Week 4: Recommendation system & report

Project Overview

Work Overview -

- Identify causes behind learner churn and low engagement.
- Build predictive models to flag at-risk learners.
- Create personalized retention strategies.
- Implement a rule-based recommendation engine.

Tools Used -

- Python, Pandas, Scikit-learn, Matplotlib
- Google Colab & Sheets
- ROC, Confusion Matrix, Feature Importance

Dataset Overview

1. Records: 8,558 learners
2. Timeframe: May 2023 - April 2025
3. Features: 24 original + 6 engineered
4. Target: Churn (Dropped Out = 1 , Others = 0)

Week 1 - Data Cleaning

Before -

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
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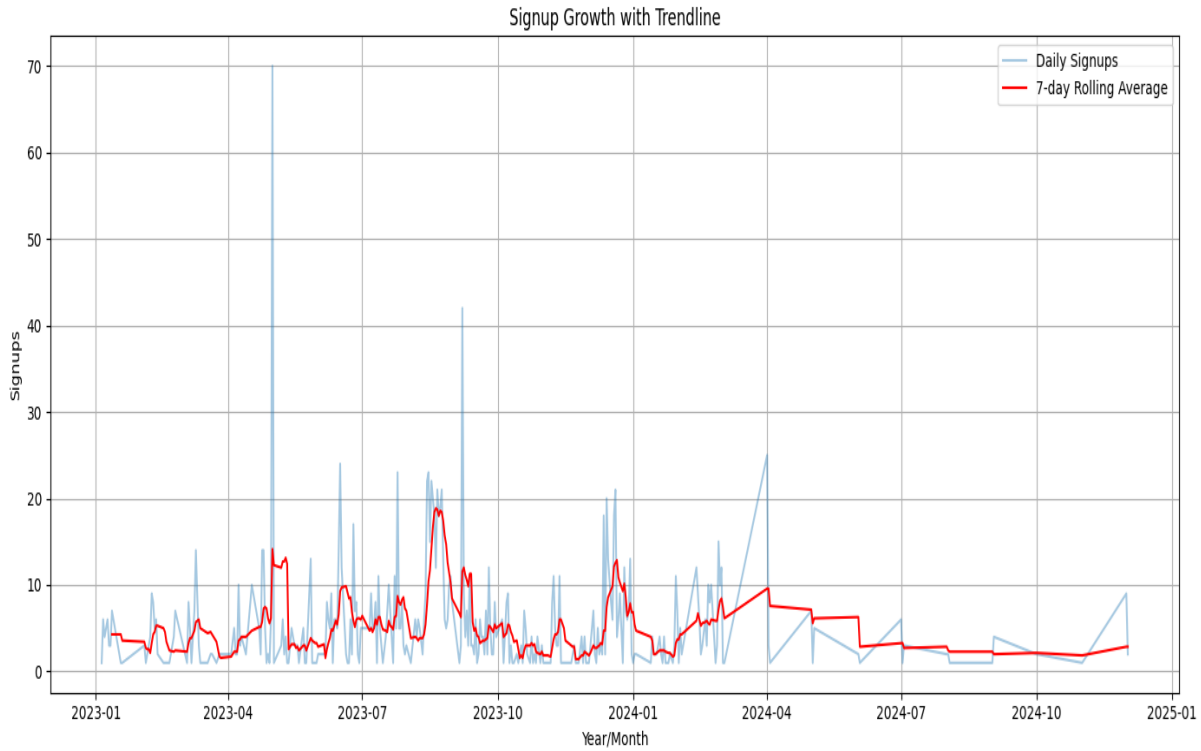
Feature Engineering Highlights

Derived Features:

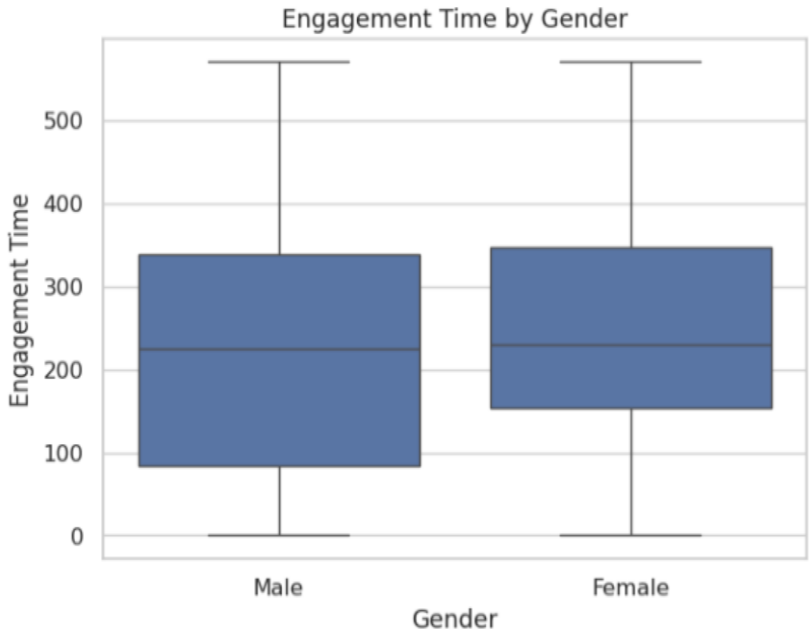
- Engagement Duration
- Time in Opportunity
- Age, Sign-Up Month, Sign-Up Year
- Gender_Male (binary)
- Encoding: One-hot encoding for categorical fields

	Q	T	U	V	W	X	Y
1	Age	Engagement Duration	Time in Opportunity	SignUp Year	SignUp Month	Gender_Encoded	
2	23	459	841	2023	6	1	
3	24	299	841	2023	1	1	
4	23	542	841	2023	9	0	
5	26	548	841	2023	8	1	
6	25	446	841	2023	6	0	
7	28	6	153	2024	2	0	
8	23	459	841	2023	5	0	
9	19	497	841	2023	7	1	
10	19	438	841	2023	3	0	
11	26	645	841	2023	11	1	
12	27	394	841	2023	4	1	
13	27	325	841	2023	1	1	
14	26	684	841	2023	8	0	
15	28	459	841	2023	6	0	

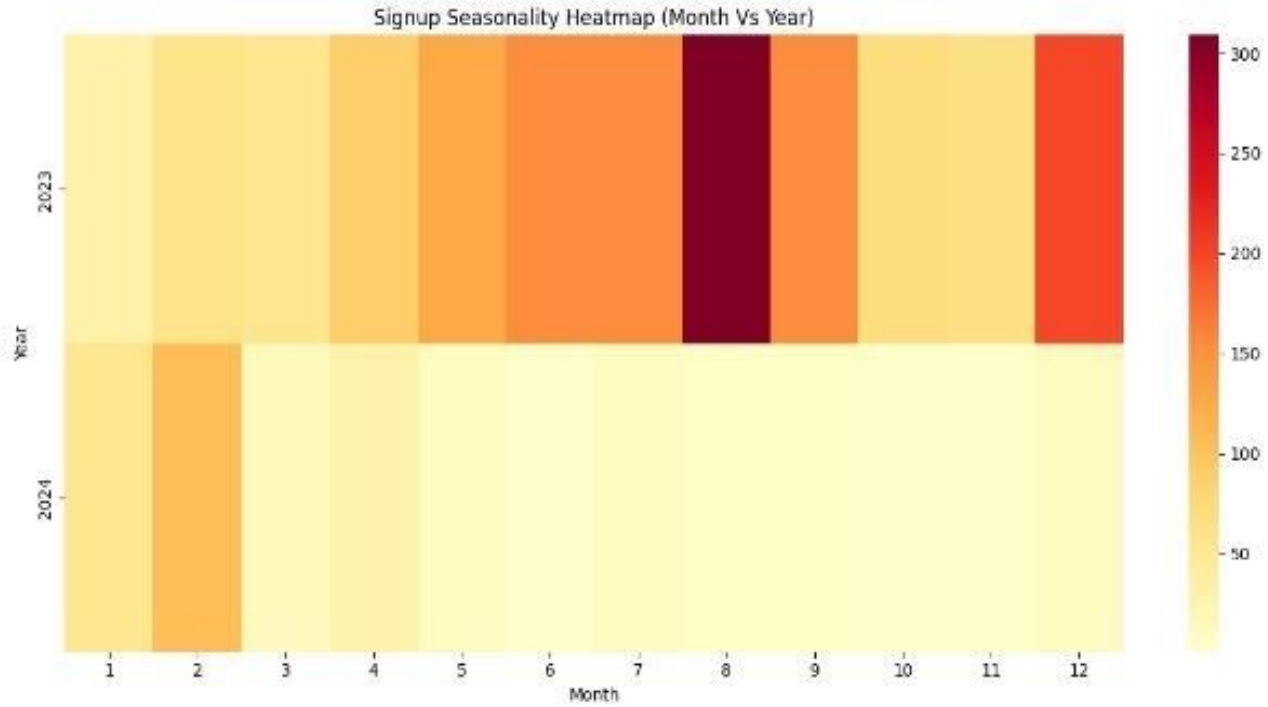
Week 2 - Exploratory Data Analysis

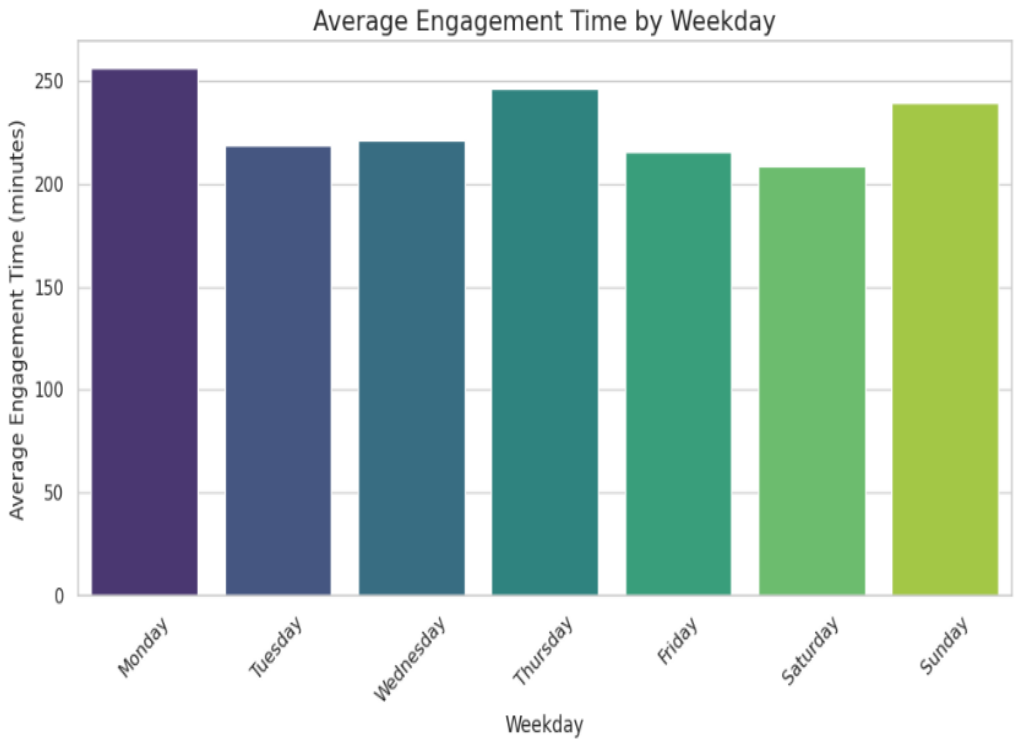


Monthly sign-ups peaked in Apr & Aug



Male learners engage longer





Higher completions on Tuesdays & Thursdays and in the months from May to August

Week 3 - Predictive Modeling

Model 1 -

Logistic Regression

Accuracy: 90.5%
Recall (Churn): 87%

Model 2 -

Random Forest Classifier

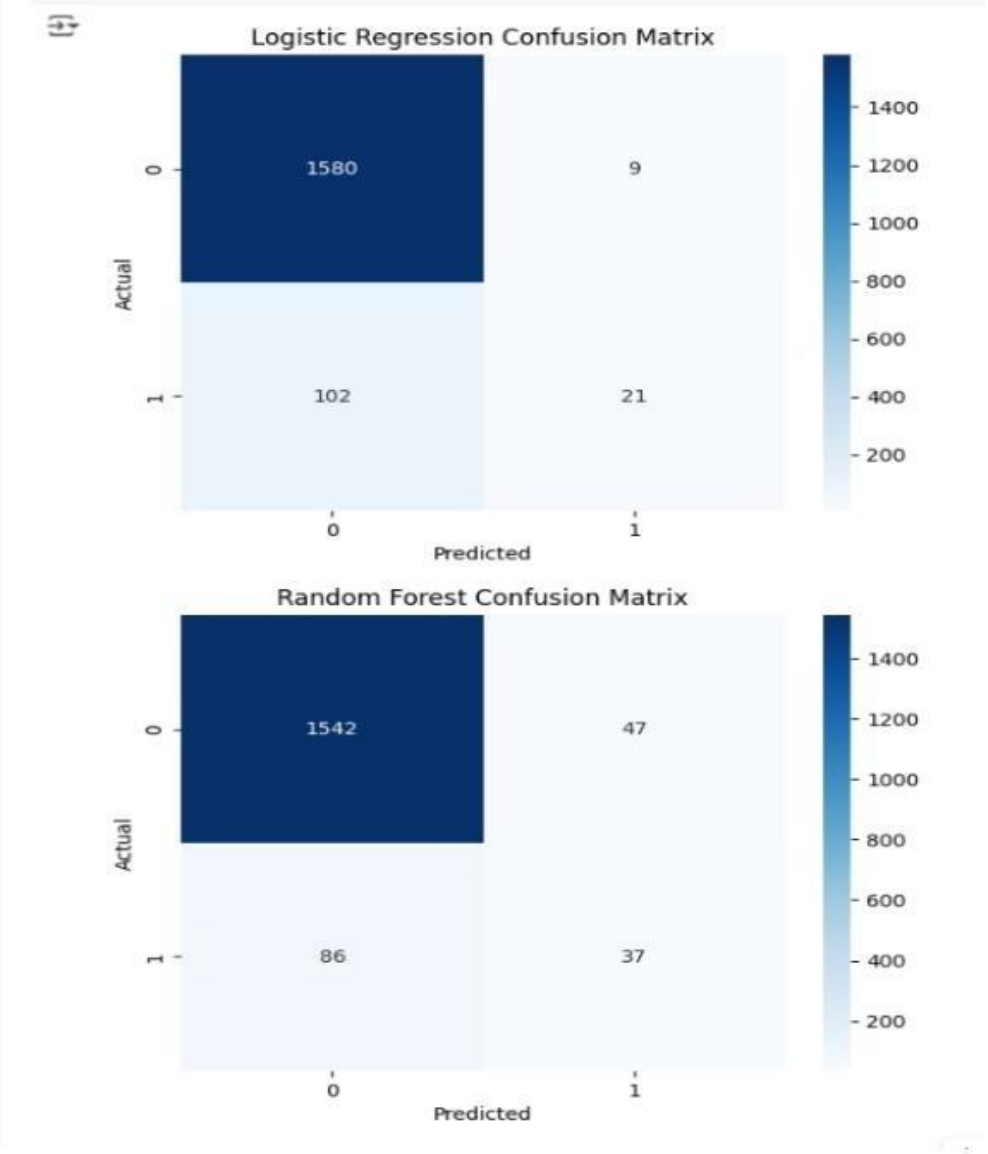
Accuracy: **92.2**%
Recall (Churn): 90%

- Model Evaluation: Stratified 80-20 split, ROC, Confusion Matrix
- Key predictors: Engagement Duration, Age, Sign-Up Timing

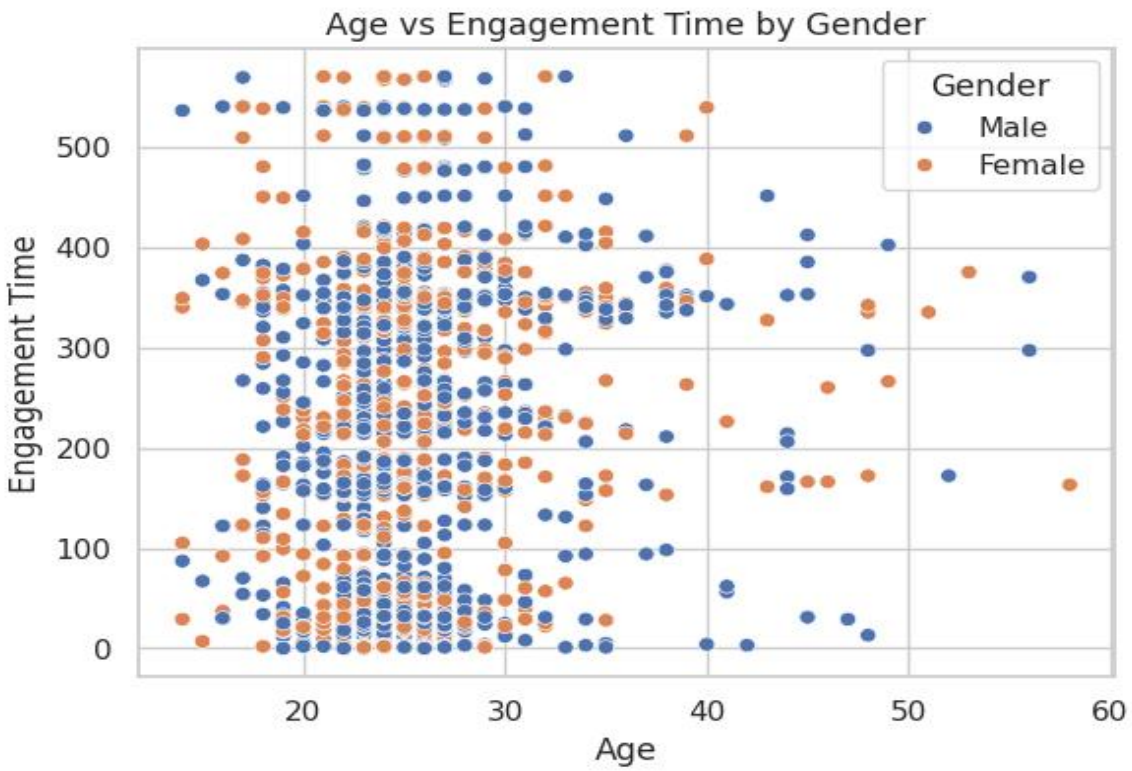
Why Random Forest Wins ?

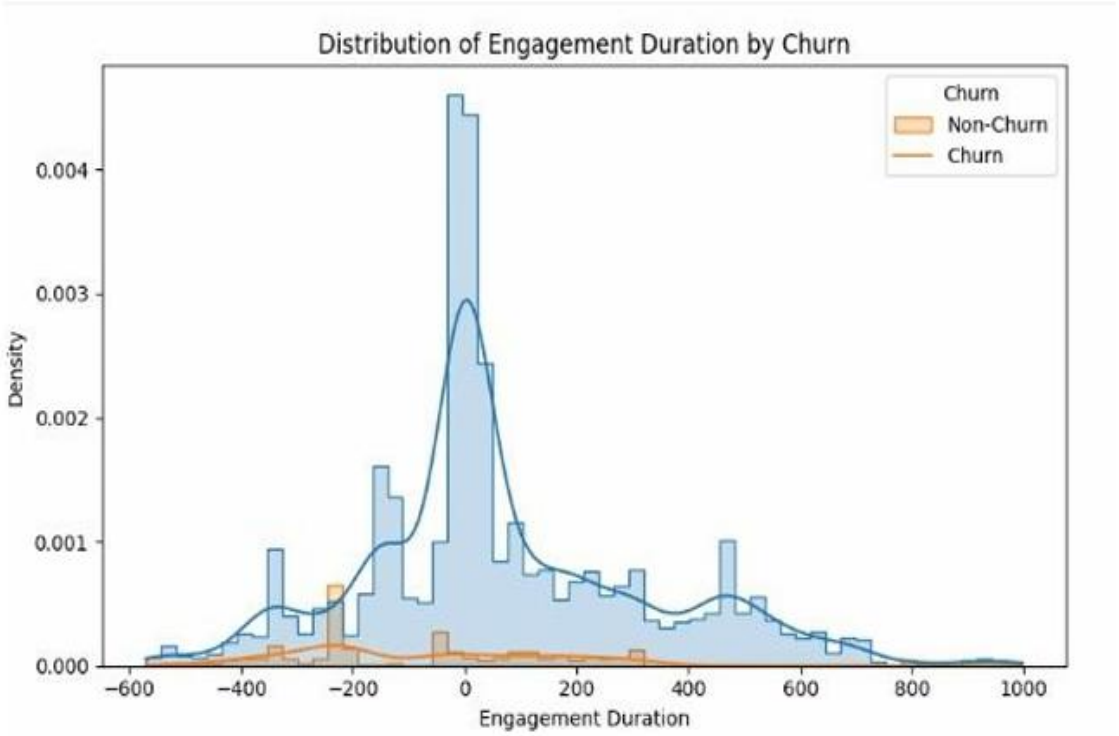
- Better at detecting churn (higher recall)
- More useful in early warning scenarios
- More balanced precision-recall trade-off

Confusion Matrix -



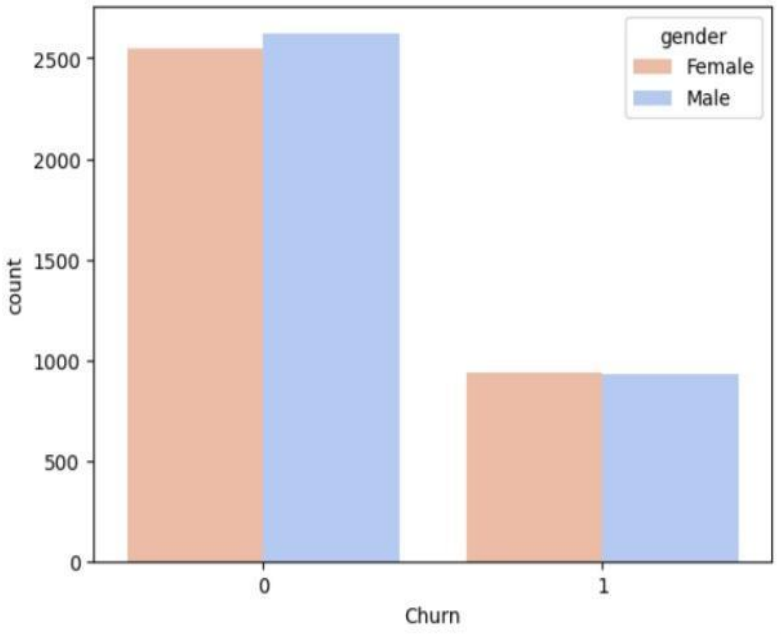
Week 3 - Churn Drivers

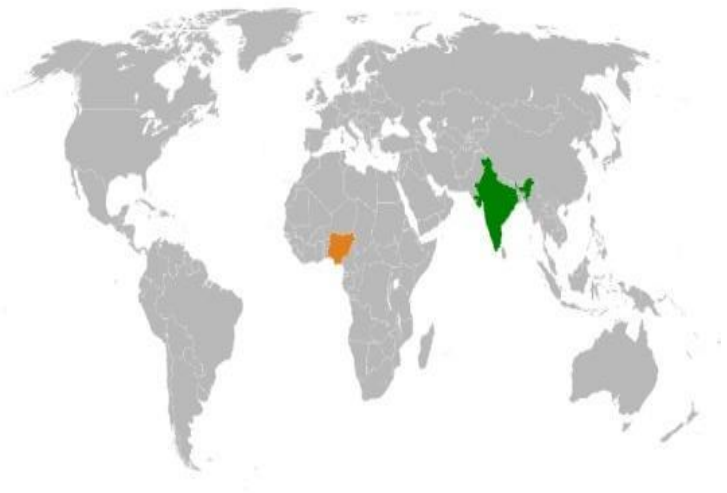




- Churners have 62% lower engagement
- Average churn rate: 7.2%
- Churners slightly younger on average

Week 3 - Demographic Patterns Results





- Male churn > female churn
- Intern participants churn more than course learners
- Highest churn in Nigeria & India

Week 4 - Recommendation System

- Rule-based engine on 5 learner attributes
- Personalized actions: mentors, tutorials, nudges
- Code -

Step 2: Define Recommendation Logic

```
[ ] def generate_recommendations(row):
        recs = []

        # Rule 1: Low engagement
        if row['Engagement Duration'] < 200:
            recs.append("Send motivational email or onboarding tutorial")
            recs.append("Assign mentor for weekly check-ins")

        # Rule 2: Younger learners
        if int(row['Age']) < 23:
            recs.append("Offer time management webinar or peer buddy system")

        # Rule 3: Internships (higher churn risk)
        if row['Opportunity Category'].lower() == 'internship':
            recs.append("Provide internship readiness workshop")

        # Rule 4: Late sign-up (short time before opportunity)
        if row['Time in Opportunity'] < 500:
            recs.append("Send early access prep materials and reminders")

        # Rule 5: Gender-based engagement (EDA showed males varied more)
        if row['Gender'] == 'Male':
            recs.append("Encourage consistent engagement via weekly nudges")

        # Fallback
        if not recs:
            recs.append("Continue with regular progress tracking")

        return "; ".join(recs)
```

Week 4 - System Summary

Learner Name	Age	Gender	Engagement Duration	Recommendation
Faria	23	Female	459	Continue with regular progress tracking
Poojitha	24	Female	299	Continue with regular progress tracking
Emmanuel	23	Male	542	Encourage consistent engagement via weekly nudges
Amrutha Varshini	26	Female	548	Continue with regular progress tracking
Vinay Varshith	25	Male	446	Encourage consistent engagement via weekly nudges

- Interpretable & scalable rules
- Attributes: engagement, age, timing, gender, opportunity
- Actionable suggestions for retention

Rule-based logic triggers tailored support:

- <200 days Engagement ⟶ Mentor assignment
- Age < 23 ⟶ Time management webinars
- Internship ⟶ Onboarding + reminders
- Male learners ⟶ Weekly motivational nudges

Recommendations Recap

- Automated onboarding
- Peer support for younger learners
- Timely reminders & resources
- Targeted outreach in high-churn regions

Expected Impact

- 20-30% reduction in churn
- Improved completion rates
- Foundation for AI-based personalization

Summary

Key Outcomes

- Built predictive churn models with up to 0.88 AUC
- Developed engagement-based intervention logic
- Proposed retention strategies based on real trends
- Expected Churn Reduction: 20–30%

Future Scope

- Real-time dashboards for churn risk
- AI-driven personalized content workflows
- Automated interventions and success coach outreach

Final Reflections

- Gained hands-on experience in real-world ML workflows
- Learned importance of recall over accuracy in social impact ML
- Built teamwork, data interpretation, and presentation skills

Let’s build platforms that help learners succeed , not just enroll.

Thank You !